Data centre water and energy: do published models match what Google reports?

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Background

Models are widely used to estimate data centre water and electricity use, but facility-level data are rare. Google discloses facility PUE, power usage effectiveness (2018 to 2025), and water (2022-2025) for named sites. We test published models against these disclosures: Lei & Masanet (2022) for PUE, Siddik et al. (2021) for water per kWh, with the EU's mandatory reporting as a benchmark.

Data and method

- Water panel: 128 site years (23, 29, 36 and 40 sites in 2022 to 2025). Sites plus ‘Other’ match Google’s printed totals.
- Fleet WUE (water usage effectiveness) = printed data centre water ÷ IT energy (electricity ÷ PUE)
- Lei & Masanet (2022): lowest PUE a US data centre could reach by climate zone. Case 1 is its best practice case.
- Siddik et al. (2021) is a published estimate of US data centre water use. It applies one national figure of 1.8 litres per kWh of facility energy to every site.
- PUE: Google's 2025 facility PUE against the Lei & Masanet Case 1 PUE for the site's climate zone; mean of campuses where a site has two
- Lei et al. (2023) samples are not location-specific, so they are not used as a test.

1. PUE: best practice is beaten at 14 of 15 sites

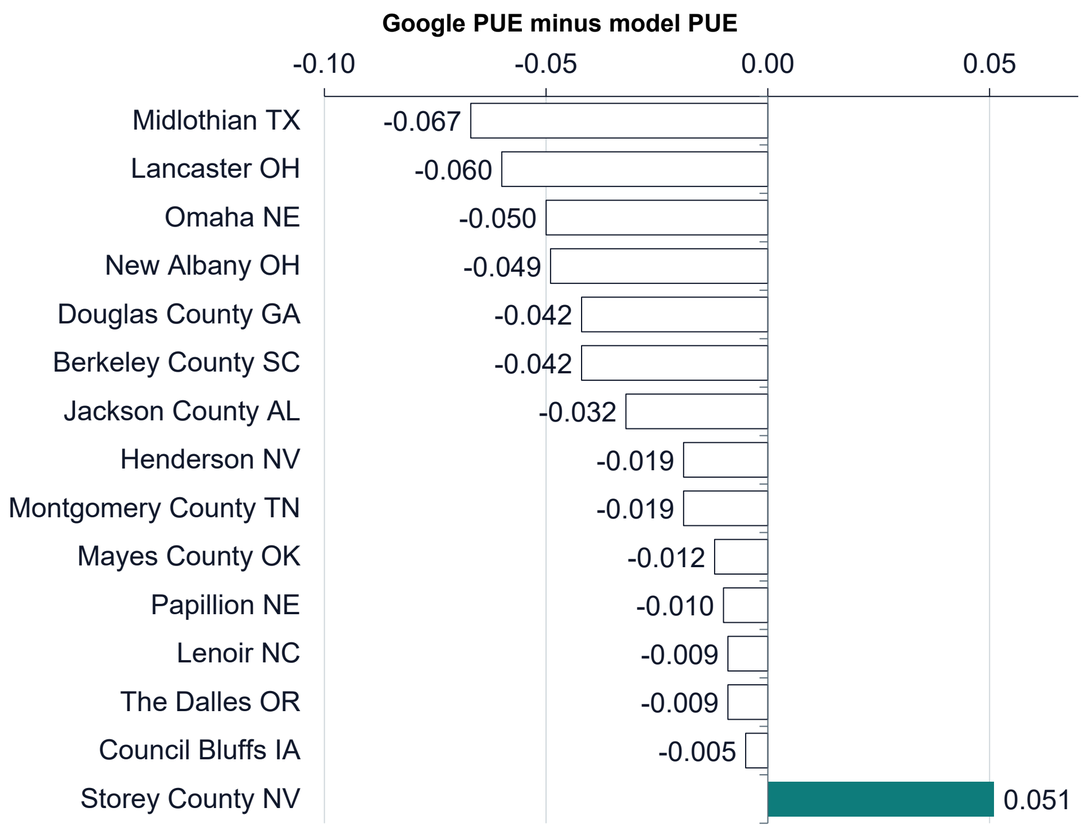


Figure 1. Google 2025 PUE minus Lei & Masanet best-practice PUE (Case 1), IECC 2021 climate zones. Below zero: Google is lower.

Three sites (New Albany, Omaha, Lancaster) also beat Case 2, a design in the model that uses about 1.7 to 1.8 L of water per IT kWh. Storey County is the one miss against Case 1. At campus level, 14 of 17 campuses beat Case 1.

Within sites, 2022 to 2025

11/16
sites where PUE improved

14/16
sites where water use rose

1.7×
median growth in water use

16 sites with PUE and water in both 2022 and 2025. Site load growth is not observed, so this is description, not an efficiency claim.

2. Water per kWh: below US constant

1.05
L per IT kWh, fleet, 2022-24

52-54%
of Siddik's 1.8 per facility kWh

1.8×
the EU 2024 average of 0.58

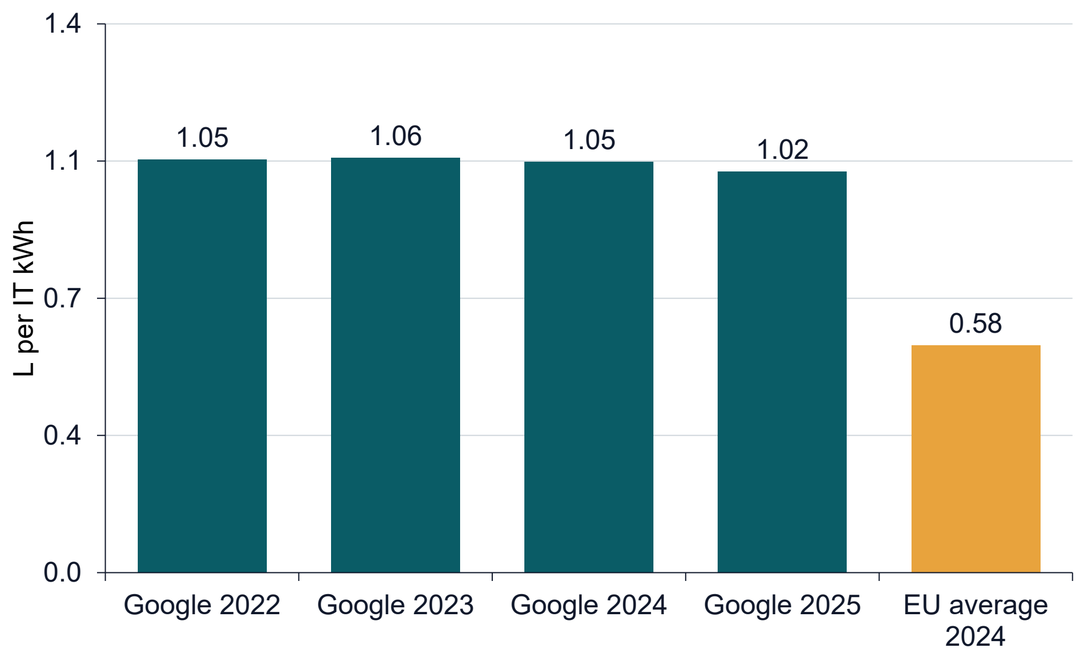


Figure 2. Google fleet WUE (water usage effectiveness, litres of water per IT kWh), 2022 to 2025, and the EU 2024 average (770 sites, Energy Efficiency Directive). Google values are printed data centre water divided by IT energy. Siddik’s 1.8 is per facility kWh, so Google’s figure is divided by PUE before comparing.

Google's water use per kWh is about half the US constant, so the constant is too high for Google's facilities. It does not make Google water-efficient: its WUE is above the EU average.

3. A plausibility check

Building footprint estimates from the Pacific Northwest National Laboratory (PNNL) IM3 model, scaled to Google’s fleet IT energy, give 0.49 to 1.44 L per kWh at 8 US sites. This range includes Google’s fleet figure of 1.05. The scaling uses one fitted factor, so it cannot test differences between sites.

Conclusions

- The PUE model is conservative for Google. The water constant is too high for Google.
- Google's PUE is at or below the models' best-practice case at 14 of 15 sites
- Google's water per facility kWh is about half Siddik's national constant, yet 1.8× the EU average
- Water use rose at 14 of 16 sites while PUE improved at 11; load growth is unobserved

Limitations

- One firm's voluntary disclosure: selection bias towards efficient operators
- Site water only; indirect water is about three-quarters of the footprint (Siddik)
- Gross consumption; Google's replenishment is not netted off
- About 10% of data centre water is in unlocated 'Other' sites
- No US facility-level energy data are public, which limits any direct test of WUE

References: Lei, N. & Masanet, E. (2022) Resources, Conservation and Recycling 182. Siddik, M.A.B., Shehabi, A. & Marston, L. (2021) Environmental Research Letters 16, 064017. Lei, N. et al. (2023) J. Phys.: Conf. Ser. 2600, 172003. Google Environmental Reports 2023-2026. European Commission, COM(2026) 500 (EU 2024 data). Charts: authors' analysis of these sources.